

Mondrian R. Syafaat

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EDUCATION

UC Merced, 08/2023 - 05/2027

Bachelors of Science: Materials Science and Engineering (Computational Emphasis)

3.3 GPA (05/2025)

- *Relevant Coursework*: Intro to Materials, Statics/Dynamics, Fluid Mechanics, Computer Aided Engineering, Engineering Economics, Chemistry II, Statistics, Vector Calculus, Physics II, Advanced Programming, Discrete Mathematics
- *In Progress*: Strength of Materials, Solid State Materials, Mechanical Behavior of Materials, Thermodynamics

PROJECTS

“Trailer with Adjustable Width” 12/2024

- Designed and modeled mechanical components and assemblies using SolidWorks, ensuring precise dimensions and functionality.
- Produced deliverables including detailed technical drawings and an animation of the mechanism to support hypothetical manufacturing/assembly processes.

“Simulated ASTM D790 Testing” 12/2025

- Used Finite Element Analysis and CAD modeling in Solidworks to characterize the stiffness-to-weight efficiency of composite sandwich structures.
- Demonstrated that sandwich construction retains ~80% of flexural stiffness while reducing core density.
- Investigated vibration damping properties by modeling viscoelastic cores (PU foam vs. Silicone gel) to optimize player comfort and reduce frame shock for future investigations.

“Maximizing Power using Carbon Fibre Orientation of a Tennis Racket” 08/2025 - Present (In Progress)

- Using Finite Element Analysis and CAD modeling in Solidworks, to analyze how orientation of carbon fibre layouts affect distribution of stress loads and energy return.

SKILLS

- Computer Aided Design and Finite Element Analysis (SolidWorks)
- 3D Printing and CNC Machining
- Metallurgical sample topography (experience with upright BX51 Microscope)
- Coding in C++, Python, JavaScript, HTML, Linux Command Line
- Leadership, collaboration, conflict resolution, and presentation skills
- Google/Microsoft Office Suite
- Fluent in English and Indonesian

PROFESSIONAL EXPERIENCE

Tennis Coach/Instructor, 06/2022 - Present (Seasonal)

iTennis (Rose Bowl Tennis Center) - Pasadena, CA

- Managing, motivating, and teaching tennis from kids as young as 5 to adults, encompassing all levels

EXTRACURRICULARS

UC Merced Tennis Club, 08/2023 - Present

- President (06/2025 - Present), Vice President (06/2024 - 05/2025), Treasurer (08/2023 - 05/2024), Coach & Team Captain (08/2023 - Present)
 - Organizing, presenting, and leading events to deliver information to members
 - Leading cabinet meetings, delegating weekly responsibilities to each cabinet member
 - Problem/resolution and relationship building between team members and school administration
 - Organizing trips, fundraisers, and community service events
 - Organizing practices for beginners and competitive members
 - Developed a meritocratic method to assemble teams based through “ELO calculation” algorithm
 - Tracked and managed finances and purchases